

Geographic Data Science

Spatial Data

Elisabetta Pietrostefani & Carmen Cabrera

Data, data, data

- Everything produces data
- New dense quantitative representations of the world
- But data is not useful, insights are

Data Science

“gathering data messaging it into a tractable form, making it tell its story and presenting that story to others”

Loukides (2011) What is Data Science?

Spatial Data Science

- A lot of new data is spatial data
- Spatial is special
- We don't want to reinvent the GIS wheel
- How do we bring both world together?

Good old spatial data (+)

Good old spatial data (+)

Traditionally, datasets used in social sciences are

- Collected for the purpose (carefully designed)
- Detailed and informative (“rich profile and portraits of the country”)
- High quality

Good old spatial data (-)

But also

- Massive enterprises (very costly)
- Coarse in resolution (to preserve privacy they need to be aggregated)
- Slow - the more detailed, the less frequent they are available

Examples

- Decennial census (census geographies)
- Longitudinal surveys
- Custom collected surveys, interviews etc.
- Economic or well-being indicators

***New Forms* of spatial data**

New Forms of spatial data

Tied into the geo-data revolution

- Accidental : created for different purposes but available for analysis as a side effect
- Very diverse in nature: resolution and quality but, potentially much more detailed in both space and time

We will look at this more in a few weeks!

Lazer & Radford (2017)

- **Digital life**: digital actions (Twitter, Facebook, Wikipedia...)
- **Digital traces**: record of digital actions (CDRs, metadata...)
- **Digitalised life**: nonintrinsically digital life in digital form (Government records, web...)

Arribas-Bel (2014)

Three levels, based on how they originate:

- **Bottom up**: “Citizens as sensors”
- **Intermediate**: Digital businesses/businesses going digital
- **Top down**: Open Government Data

Opportunities (Lazer & Radford, 2017)

- Massive, passive
- Nowcasting
- Data on social systems
- Natural and field experiments (“always-on” observatory of human behaviour)
- Making big data small

Challenges (Arribas-Bel, 2014)

- Bias
- Technical barriers
- Methodological “mismatch”

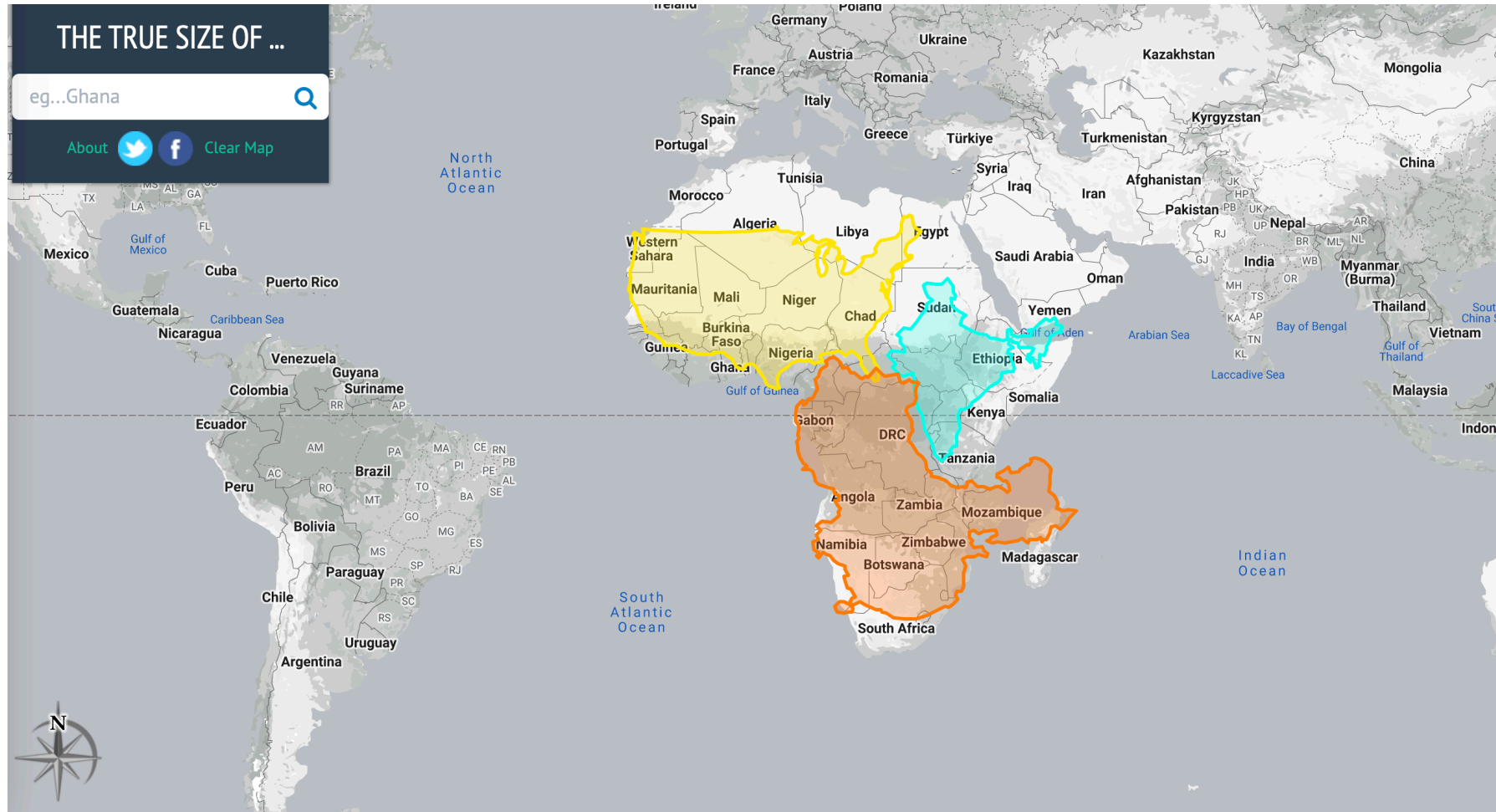
All maps are wrong

All maps are wrong



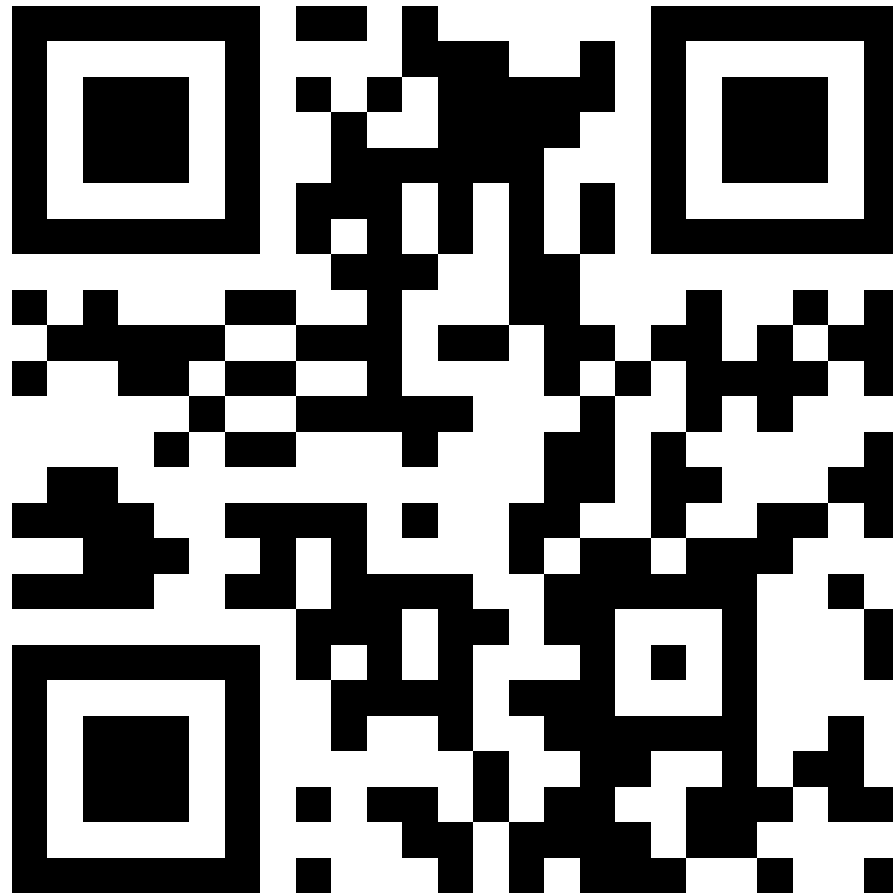
Tell your neighbour how maps can lie.

All maps are wrong



Still not convinced?

Scan to play with thetruesize.com — drag any country around the globe and watch its true size change.



Breaking: *Correcting* the map

On 4 September 2026, the UN General Assembly voted **164–1** (6 abstentions) to adopt “Correct the Map” — encouraging a shift from Mercator to the **Equal Earth** projection.

Non-binding. Doesn't ban Mercator for navigation. Doesn't redraw a single border.

UN News · The Guardian · Al Jazeera · CBC · also widely reported by the BBC

Not everyone agreed

164 in favour, **1 against** (the United States), 6 abstentions.

US representative: the resolution is “*the reason this institution is losing its credibility*” — the Assembly shouldn’t be “*debating map projects from the 16th century.*”

Togo’s FM Robert Dussey: maps shape “*education, imagination and collective perception.*”

Why it matters

On Mercator: Greenland $\approx$ Africa in size

In reality, Africa is about **14× larger** than Greenland.

Mercator is **conformal** (preserves shape/angle) → area gets distorted, worse toward the poles.

Equal Earth is **equal-area** (preserves area) → shape gets distorted instead.

The EPSG codes

(the news coverage never mentions them — here’s the definitive list)

CRS	EPSG
Equal Earth (projection method)	1078
WGS 84 / Equal Earth Greenwich	8857
WGS 84 / Equal Earth Americas	8858
WGS 84 / Equal Earth Asia-Pacific	8859
World Mercator	3395
Web Mercator (Google Maps, OSM)	3857

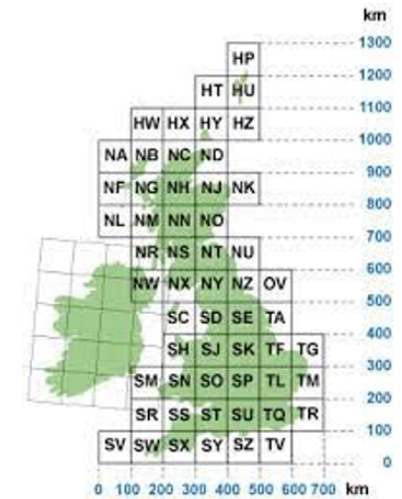
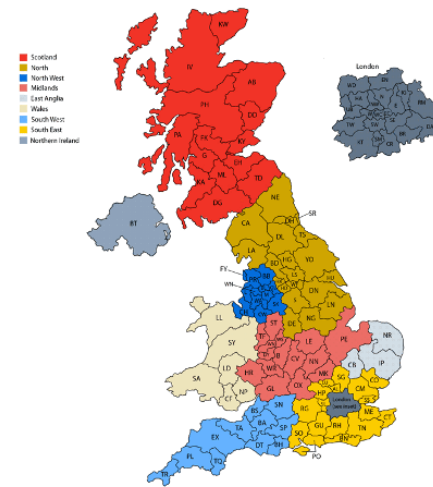
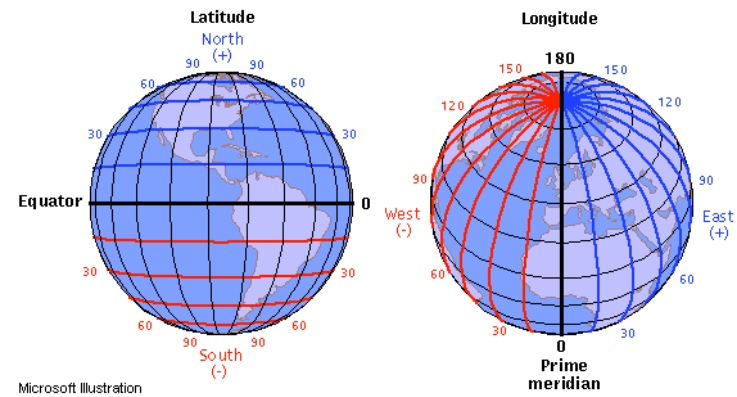
Try it yourself

Al Jazeera: guess the real size of each country

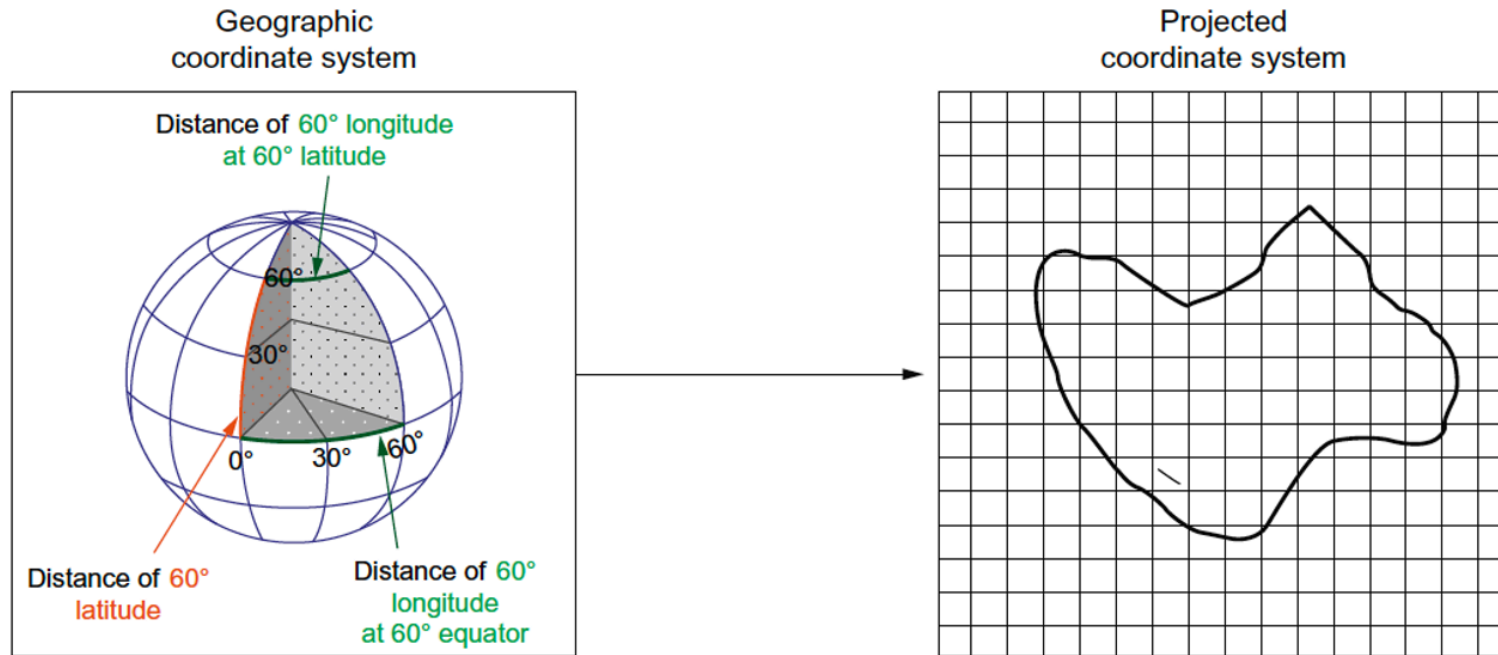
Pick which country is bigger, watch it morph from Mercator to Equal Earth, see how close you were.



Linking Spatial Info



Linking Spatial Info



Coordinate Reference Systems

Coordinate reference systems provide the context of coordinates:

- They tell whether the coordinates are **ellipsoidal (angles)**, or derived, **projected (Cartesian)** coordinates
- In case they are projected, they detail the kind of projection used, so that the underlying ellipsoidal coordinates can be recovered

Coordinate Reference Systems

- Convert between projected and unprojected, or to another projection
- Transform from one datum to another
- Combine the coordinates with any other coordinates that have a coordinate reference system

**Have a better look at the concepts
page**



Questions



Geographic Data Science by [Elisabetta Pietrostefani](#) is licensed under a [Creative Commons Attribution-ShareAlike 4.0 International License](#).